EnergyMind Tunisia – Goal 1

Anomaly and Bias Detection for a Trustworthy Smart-Metered Grid

Phase 1 technical note: types of anomalies, families of detectors, what we use, and why.
All figures are generated from code (data files + TikZ/PGFPlots).

In one paragraph. Our system must decide, for every suspicious reading, whether the *grid* did something unusual or the *data* is wrong, because the two need opposite responses. No single detector can do this. We therefore combine (i) physics and statistical rules, which are auditable, (ii) three unsupervised machine-learning detectors trained only on clean data, which are sensitive to subtle patterns, and (iii) a cross-sensor consistency reasoner that separates genuine events from data-quality problems. Section 5 tests the detection layers on node N1 of the EnergyMind dataset and reports, per fault type, what they catch and what they still miss.

1 The Problem: a Suspicious Number Has Two Possible Meanings

An anomaly is a pattern that deviates from expected behaviour [1]. In our prototype node (N1 of the EnergyMind dataset: one transformer, 30 smart meters on 3 low-voltage feeders of 10, three homes with rooftop PV, readings every 30 minutes, each feeder metered at its head) a surprising value can come from the **grid** or from the **data chain**, and the right response differs: a real event needs operational action, a faulty or falsified sensor needs to be distrusted and repaired. A further difficulty is that an informed attacker can craft false data that passes standard bad-data checks [3], so one test is never enough.

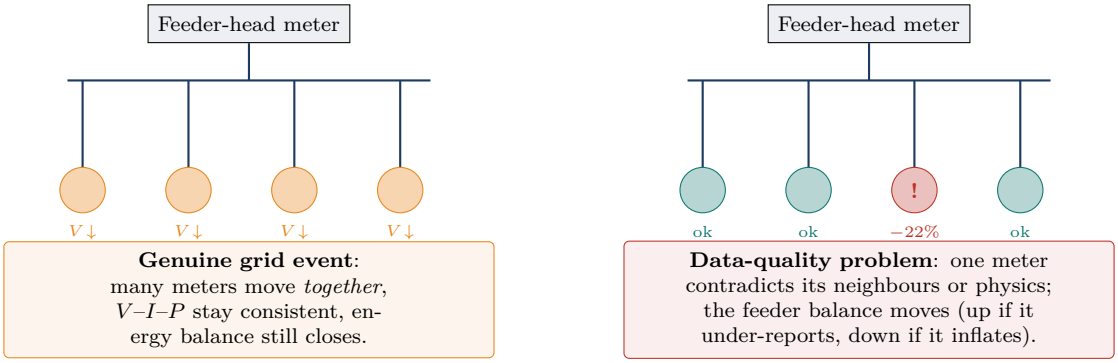


Figure 1: The core question. Left: a real event is visible on many sensors and obeys physics. Right: a faulty or falsified channel disagrees with redundant measurements. This cross-sensor logic is the backbone of our design.

2 Types of Anomalies

The literature classifies anomalies by how they relate to context [1]. We use that classification and map it onto grid failures (Figure 2, Table 1).

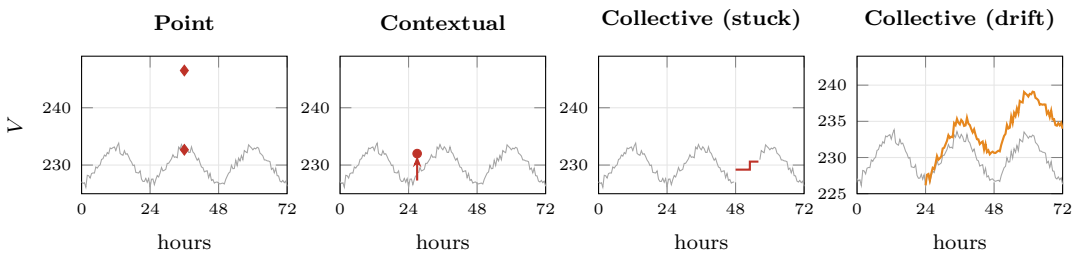


Figure 2: Illustrative voltage traces (synthetic, 30-min samples as in the dataset). Grey: normal behaviour. Red/amber: the anomaly. A *point* anomaly is one odd value; a *contextual* anomaly is a value that is normal in general but wrong for its time of day; *collective* anomalies are odd only as a sequence (a frozen sensor, a slow drift).

Project anomaly	**Type [1]**	**Signature in the data**	**First line of defence**
Impossible value (spike, sign or unit error, saturation)	Point	$E < 0$, $E = 99.999$, $\times 1000$, $\times 8$–40	Range rules
Stuck meter	Collective	Every channel frozen (E *and* V)	Flat-line rule
Data-only injection (scale, evasion, inflate, coordinated)	Contextual	$E \neq VI \cos\varphi \, \Delta t$	Physics rule (skipped while exporting)
Replay	Collective	Same half-hour one week earlier	Profile residual, ML ensemble
Voltage sensor offset	Collective	Meter V persistently *above* feeder-head V	V-vs-head rule, CUSUM
Sensor bias, gain drift (± 2–14%)	Collective	V, I, E scaled together: physics holds	Feeder balance (feeder level), profile, ML
Physical bypass, zeroing	Contextual	Energy *and* current reduced: physics holds	Feeder balance, zero-run, ML
Clock shift	Collective	All channels delayed 1–3 h	Lag correlation with profile
Genuine outage, sag, swell, surge, cable fault	(not an error)	Many meters, physics holds; $V = 0$ on supply loss	Flagged here; classified by the reasoner

Table 1: From anomaly types to the fault types of the EnergyMind dataset. Genuine events are *expected* to trigger the detectors; the reasoner (Section 5) must not call them bad data. Missing readings (communication loss) are not scored: they are passed on as "no data".

3 Families of Detection Methods and Our Choice

Methods can be organised by *what knowledge they use* and by *how many labels they need* [1, 2]. Because labelled faults are rare in grids, we are driven toward the semi-supervised and unsupervised regimes: learn what "normal" looks like from clean data, and flag deviations (Figure 3).

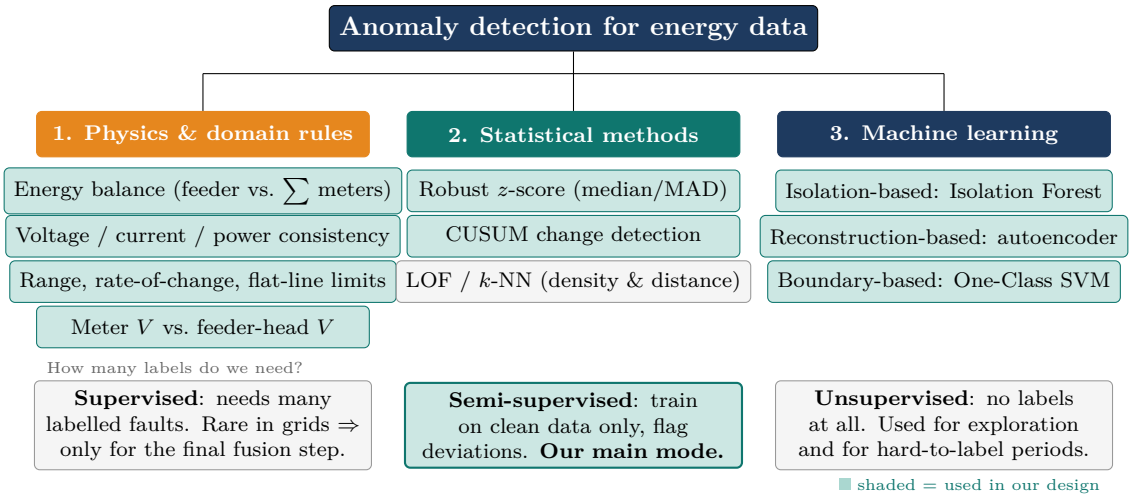


Figure 3: Taxonomy of detectors. Teal boxes are the methods we use (detailed in Section 4). Hybrid physics + learning follows the argument that high-stakes decisions should rest on inspectable models where possible [4].

4 The Methods, One by One

4.1 Statistical Outlier Methods

Robust z-score. The classic $z = (x - \bar{x})/s$ is distorted by the very outliers it seeks, because mean and standard deviation have a 0% breakdown point. Replacing them by the median and the scaled median absolute deviation gives

$$z_i^{\text{rob}} = \frac{x_i - \text{med}(x)}{1.4826 \cdot \text{MAD}(x)}, \qquad \text{MAD} = \text{med}_i |x_i - \text{med}(x)|, \tag{1}$$

which tolerates up to 50% contamination [5, 6]. **CUSUM** [7] accumulates small deviations, $S_t = \max(0,\, S_{t-1} + (x_t - \mu_0)/\sigma - k)$, and alarms when $S_t > h$; it is the natural tool for slow drift. **LOF** [8] compares a point's local density with that of its neighbours, so it handles clusters of different density.

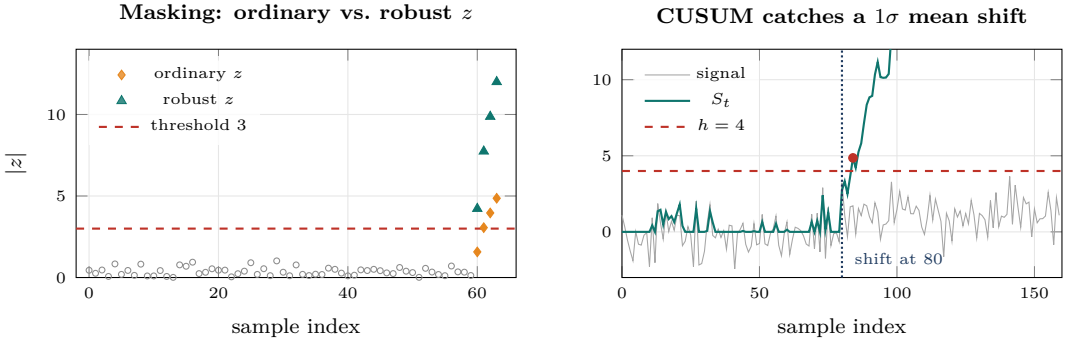


Figure 4: Left: four injected outliers among 60 normal samples. Ordinary z gives 1.6, 3.1, 4.0, 4.9, so 1 of them hides below the threshold (masking); robust z gives 4.2, 7.7, 9.9, 12.0, so all four are flagged. Right: CUSUM ($k = 0.5$, $h = 4$) alarms at sample 84, 4 samples after the shift, although the raw signal stays mostly within $\pm 3\sigma$. Both from our `gen.py`, synthetic data, fixed seed.

Role in our design. Cheap, transparent first line and the baseline every ML component must beat. Robust z is applied to residuals from each meter's own 28-day half-hourly profile, so the

evening peak, weekends, Ramadan and heat-wave days are not mistaken for anomalies. CUSUM runs on two slow signals: the meter's voltage relative to its feeder head (a sensor offset) and the feeder's daily energy-balance gap (a bias or bypass somewhere on the feeder).

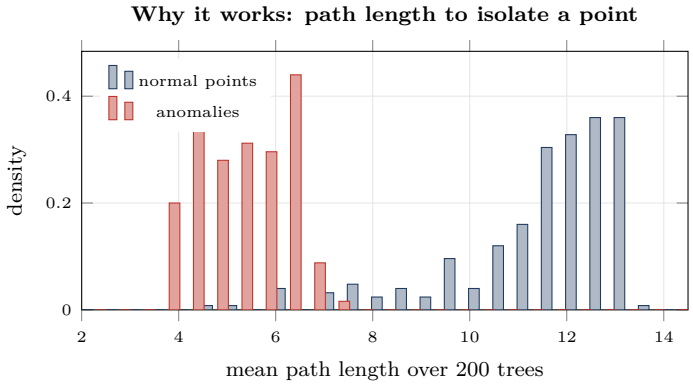
The physics rules and what they cannot see. The rule $E \approx V I \cos\varphi \, \Delta t$ exposes *data-only* injection, which edits the energy field and leaves V and I untouched. It is skipped while a meter exports PV energy (import is then 0 while current flows; without the gate it fires on 39% of the clean readings of N1's PV homes). It is *blind* to sensor bias, gain drift and physical bypass, which scale energy and current together, so those cases rest on the feeder balance and on the meter's own history. A meter cannot read a higher voltage than the head of its feeder, because voltage only drops along the line; a persistent excess (24-h median above +1.5 V, export readings excluded) is a voltage-sensor offset.

4.2 Isolation Forest

Liu, Ting and Zhou [9] observed that anomalies are few and different, so random axis-parallel cuts isolate them in fewer splits. A tree is grown by repeatedly picking a random feature and a random split value; the path length $h(x)$ to isolate x is short for anomalies. With subsample size ψ the score is

$$s(x,\psi) = 2^{-E[h(x)]/c(\psi)}, \qquad c(\psi) = 2H(\psi-1) - \frac{2(\psi-1)}{\psi}, \qquad H(i) \approx \ln i + 0.5772, \tag{2}$$

so $s \to 1$ for anomalies and $s < 0.5$ for normal points. It is linear in time, needs no labels and no distance computations. The Extended Isolation Forest uses randomly oriented cuts and removes the axis-parallel artefacts visible in score maps [10], which matters for correlated quantities such as V and I.



Reading the plot. Normal points sit deep inside dense regions, so many random cuts are needed to separate them (mean ≈ 11.6). Anomalies are isolated after only 5.2 cuts on average. A short average path means a high score s. Same kind of 2-D cloud as Figure 6.

Figure 5: Average isolation path length from the fitted Isolation Forest (200 trees; 250 normal points and 250 anomalies drawn outside the dense core).

4.3 One-Class SVM

Schölkopf et al. [11] separate the data from the origin in a kernel feature space with maximum margin:

$$\min_{w,\xi,\rho} \ \tfrac{1}{2}\|w\|^2 + \frac{1}{\nu n}\sum_i \xi_i - \rho \quad \text{s.t.} \quad \langle w, \phi(x_i)\rangle \geq \rho - \xi_i, \quad \xi_i \geq 0, \tag{3}$$

and $f(x) = \operatorname{sgn}(\langle w, \phi(x)\rangle - \rho)$ with an RBF kernel. The parameter ν upper-bounds the fraction of training points treated as outliers. The closely related SVDD [12] encloses the data in a minimal hypersphere.

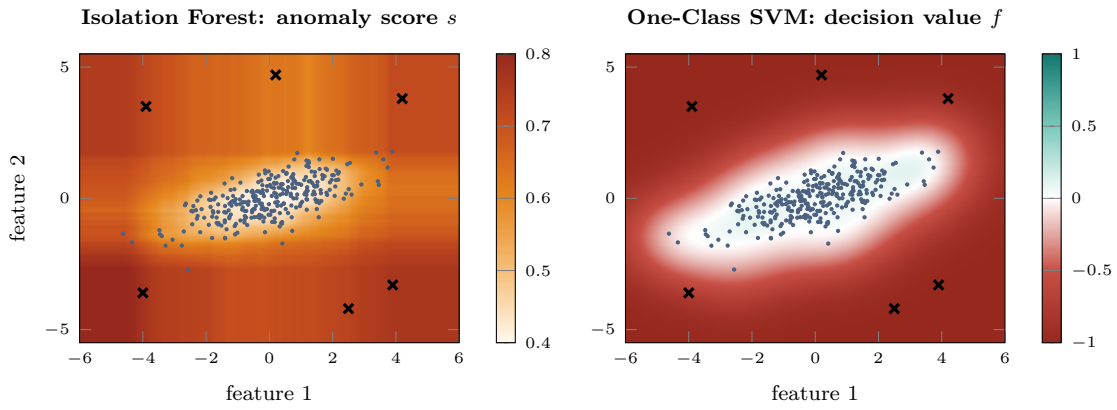


Figure 6: Two detectors fitted by `scikit-learn` [16] on the same correlated 2-D "normal" cloud (blue dots); crosses are injected outliers. Isolation Forest (200 trees, $\psi = 256$) scores the outliers 0.63–0.77 against a mean of 0.46 for normal points. The One-Class SVM ($\nu = 0.05$, RBF; decision value scaled to $[-1, 1]$) draws a smooth boundary (white) around the data. Their different inductive biases are why disagreement carries information.

Role in our design. Both run on daily window features of each meter (48 readings): profile residual (mean, maximum, spread), change of the meter's share against its feeder neighbours, physics ratio, V minus feeder-head V, share of zero readings, longest frozen run, lag gain against the profile (clock shift), similarity to the same day a week earlier (replay), distortion and minimum/maximum voltage. Isolation Forest is our fast first ML pass. OCSVM is trained on a subsample (it scales poorly) as a second, differently-biased opinion. When they disagree, confidence drops and the case is routed to a human.

4.4 Autoencoders

An autoencoder learns to compress and reconstruct normal data through a narrow code. Trained on clean data only, it reconstructs anomalies badly, so the reconstruction error is the anomaly score [13]:

$$e(x) = \|x - g(f(x))\|^2 = \textstyle\sum_j e_j, \qquad e_j = (x_j - \hat{x}_j)^2. \tag{4}$$

LSTM encoder–decoders extend this to multi-sensor time series [14]; surveys of deep detectors discuss when they help and where they over-generalise and reconstruct anomalies too well [2]. Crucially for us, the *per-channel* error e_j points at *which sensor* looks wrong, and it can be fed to Shapley-value explanations [15].

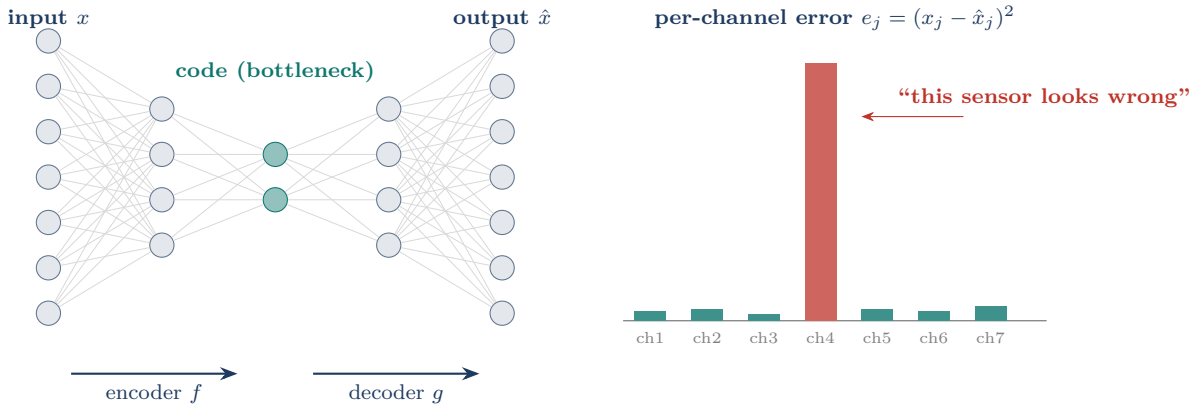


Figure 7: Autoencoder principle (left) and the per-channel reconstruction error that localises the faulty sensor (right; illustrative values).

Role in our design. Trained on clean windows only; captures correlated-sensor inconsistencies and subtle waveform changes that thresholds miss, and gives channel-level evidence for the explanation sentence ("channel 4 deviates most").

4.5 The Methods at a Glance

Method	Labels	Run cost	Explainable	Strong on	Weak on
Rules / physics	none	very low	by design	Impossible values, data-only injection, voltage offset	Bias and bypass (physics still holds)
Robust z / CUSUM	none	very low	high	Spikes; persistent offsets	Seasonal baselines (feeder balance)
Isolation Forest	none	low	medium (TreeSHAP)	Global outliers, many features	Shifts inside the normal range
One-Class SVM	none	high on big data	low (post-hoc)	Smooth boundary on compact data	Scale; kernel tuning
Autoencoder	none	medium (training)	medium (per channel)	Correlated-sensor, waveform change	Can reconstruct anomalies too well

Table 2: Qualitative comparison, our assessment from the cited literature [1, 2] and the experiment in Section 5.1. Green = favourable, amber = acceptable, red = costly.

5 Putting It Together: Our Hybrid Detection Stack

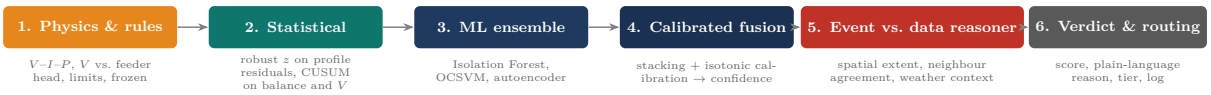


Figure 8: The detection pipeline. Rules can veto or force escalation; ML scores add sensitivity; the reasoner decides *what kind* of unusual it is.

How the verdict is reached. Layers 1–3 answer "is something unusual?" and are *meant* to fire on genuine events too; layer 5 answers "what kind?" from consistency features (Table 3). Layers 5–6 are specified in the Goal 2 and Goal 3 notes; this note covers layers 1–4.

Evidence	Genuine event	Data-quality problem
Spatial extent (meters affected)	Many	One or a few
Physical consistency (V, I, P)	Holds	Violated (data-only injection) or holds (bias, bypass)
Feeder energy balance	Unchanged	Opens (under-reporting) or goes negative (inflation)
Supply	$V = 0$ on many meters (outage)	$V > 0$
Agreement with correlated neighbours	Yes	Contradicts them
Temporal shape	Physical dynamics	Step, frozen, replayed
Weather / time-of-day context	Explains it	Does not explain it

Table 3: Consistency features used by the event-versus-data reasoner.

5.1 Results on Node N1 of the Challenge Dataset

Set-up. Unit of analysis: one meter-day (48 readings) of node N1, from 8 January (the profiles need one week of history) to 31 December, excluding days that are mostly missing: 10,715 meter-days, of which 9,494 are clean and the rest belong to 74 labelled events. Every detector is trained on clean meter-days only; labels are used only to score. Thresholds give a 1% false-positive rate on held-out clean days (the stack: rules, or any learned detector at 0.25%). Everything is out-of-fold over 5 folds, with clean days grouped by meter-week and anomalous days grouped by event, so no event is split. A time split is not possible: in N1, November–December contain no falsification and no sensor-bias event. An event counts as *detected* when at least 20% of its meter-days are flagged (any flag for events of five days or less), so a long event is not "caught" by a chance alarm. Code: `n1_experiment.py`.

Detector	**Genuine event** days	events	**Meter fault** days	events	**Falsification** days	events	**False positives**
Rules (layer 1)	42	17/36	29	18/30	66	5/8	0.0%
Robust z	1	3/36	26	12/30	61	5/8	1.1%
CUSUM	1	4/36	12	2/30	6	0/8	2.1%
Isolation Forest	3	11/36	7	15/30	9	2/8	0.9%
One-Class SVM	74	30/36	31	25/30	67	6/8	1.3%
Autoencoder	74	30/36	28	22/30	64	5/8	1.1%
Detection stack	74	30/36	31	23/30	69	6/8	2.3%

Table 4: Node N1, out-of-fold. "days": share (%) of the class's meter-days flagged (darker = higher). "events": events detected / events. False positives: share of clean meter-days flagged. Genuine events: outage, load shedding, sag, swell, surge, cable fault, transformer; meter faults: corruption and sensor faults; falsification: the 8 manipulation types. Simulated data.

Event (N1)	**Feeder**	**Start**	**Days**	**Detected by**	**First flag**	**Days flagged**	**Balance alarm**
`partial_reduction_bypass`	F2	04 Feb	60	–	day 8	18%	day 13
`selective_bypass`	F2	09 Feb	65	–	day 3	5%	day 8
`meter_zeroing`	F1	11 May	20	IF, SVM	day 0	22%	–
`fdi_scale`	F3	02 May	63	R, z, SVM, AE	day 1	98%	day 14
`fdi_evasion`	F2	25 Apr	35	R, z, IF, SVM, AE	day 0	100%	day 9
`fdi_inflate`	F3	12 Sep	33	R, z, SVM, AE	day 1	97%	day 13
`fdi_replay`	F2	09 Aug	29	R, z, SVM, AE	day 0	100%	–
`coordinated_fdi`	F1	14 Oct	26	R, z, SVM, AE	day 0	100%	day 6
`bias_step`	F1	05 Jun	58	–	day 2	5%	day 14
`gain_drift`	F2	19 Apr	66	–	–	0%	day 15
`gain_drift`	F2	18 May	65	–	day 42	2%	day 16
`clock_shift`	F1	23 Feb	15	–	–	0%	–
`voltage_offset_drift`	F1	10 Apr	64	R, z, C, SVM, AE	day 0	100%	day 3
`stuck_value`	F1	12 Feb	3	R, z, IF, SVM, AE	day 0	100%	–
`stuck_value`	F2	28 Jun	1	R, z, IF, SVM, AE	day 0	100%	–
`stuck_value`	F3	10 Nov	2	–	–	0%	–

Table 5: The 16 falsification and sensor-fault events of N1, one by one (most types occur once, so percentages per type would be meaningless). R = rules, z = robust z, C = CUSUM, IF, SVM, AE = autoencoder. First flag and days flagged refer to the whole stack. Balance alarm: first day the CUSUM on the feeder's daily energy-balance gap fires (it is in alarm on 5.2% of clean feeder-days); it names a feeder, not a meter.

What the results show.

- **The stack is usable as a first filter.** It flags 2.3% of clean meter-days. The physics and domain rules alone flag none, yet they catch every data-only injection, the voltage offset and two of the three stuck meters from the first or second day (the third froze for 4 hours across midnight, below the 3-hour-per-day limit on either side).
- **No detector wins everywhere.** The One-Class SVM and the autoencoder carry the learned part (about three quarters of genuine-event days, most meter-fault events). Robust z is strong on falsification days but blind to genuine events. Isolation Forest is weak at this false-positive rate: most anomalies here shift values *inside* the normal range rather than isolate them.
- **The hard cases are the physically consistent ones.** Partial and selective bypass, the constant bias and both gain drifts keep V, I and E consistent and are flagged on at most 18% of their days, as is a clock shift. This matches the difficulty ranking published with the dataset.
- **The feeder balance sees them, but only as a feeder.** Its CUSUM fires 8–16 days after the start of both bypasses, the bias and both drifts. Pointing to the right meter fails: a household's daily share of its feeder varies far more than a 7–14% bias, and the gap itself moves: about 0.6 points of day-to-day noise on clean days, plus a spring–summer drop when PV homes export. Phase 2 will model the expected gap from feeder load, PV export and temperature, run CUSUM on the residual, and attribute alarms with longer windows.
- **Limits.** Simulated data from one generator; most falsification and sensor types occur once in N1, so these are case results, not rates. Feeder-level references use clean feeder-days.

5.2 Output to the Explanation Layer (Goal 2)

For every flagged meter-day, the detection layer passes on: the stack decision and each detector's score; which rules fired, with the measured value and threshold; the neighbourhood support (share of the feeder's 10 meters flagged the same day); the feeder-balance alarm state; and the autoencoder's per-feature reconstruction error, which tells Goal 2 which channel looks wrong.

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